

Weather Forecasting Using Machine Learning

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CSE 1021 Introduction to Problem Solving and Programming

Introduction

Beginner friendly Python ML application

Uses historical weather observations

Estimates temperatures from weather features

Educational project , not professional forecasting

Problem Statement

Weather changes with multiple environmental factors

Goal :demonstrate a simple supervised ML approach

Input: Humidity , pressure , wind speed , rainfall

Output: estimated temperature

Objectives

Load and clean CSV data

Analyze and visualize weather

Train Linear Regression

Predict temperature for new input

Requirements

Functional: Data input, processing, visualization, prediction, evaluation

Non-Functional: Performance, usability, reliability, maintainability, resource efficiency, error handling

System Architecture

User---->Interface----->CSV Dataset----->Preprocessing----->Visualization/
Model Training---->Evaluation----->Prediction

Machine Learning Algorithm

Linear Regression

Temperature= $b_0 + b_1(\text{Humidity}) + b_2(\text{Pressure}) + b_3(\text{Wind Speed}) + b_4(\text{Rainfall})$

80%training/20% testing

Implementation

Pandas for data processing

Scikit-learn for Linear Regression

Matplotlib for plots

Modular files : processor , model, visualization, predictor, main

Pytest for testing

Future Enhancements

Real documented weather dataset

Real time weather API

Compare multiple ML methods

GUI/web application

Additional predictions

Time-series models

Conclusion

Demonstrates complete beginner ML workflow

Modular , testable and documented

Shows data processing, regression, evaluation and prediction

Suitable as an academic learning project